

```
1 #ifndef LL
2 #define LL
3
4 #include <stdio.h>
5 #include <stdlib.h>
6 #include <assert.h>
7 #include <stdbool.h>
8
9 typedef struct node {
10     int val;
11     struct node* next;
12 } node_t;
13
14 typedef struct list {
15     node_t* head;
16     node_t* tail;
17     int length;
18 } list_t;
19
20 node_t* new_node(int n);
21
22 list_t* create_empty_list();
23 list_t* create_example(int n);
24 list_t* create_single(int n);
25
26 list_t* deep_copy(list_t* l);
27
28 bool is_empty(list_t* linked);
29
30 void show_list(list_t* linked);
31 void show_values(list_t* linked);
32
33 void push(list_t* linked, int n);
34
35 void append(list_t* linked, int n);
36
37 void insert(list_t* linked, int index, int x);
38
39 int pop(list_t* linked);
40
41 node_t* free_get_next(node_t* node);
42 void free_list(list_t* linked);
43
44 #endif
```